7 Reference

FULL

What gets installed on your computer

Quick version

No time to read it all? This is the short version. The full guide is below.

- Every program the garden setup can install, why it is there, and its official page. Nothing else is installed.
- Three groups: **Necessary** (the install does them), **Optional** (the install asks yes or no) and **Advanced optional** (later, from menu `M`).
- Necessary includes Ubuntu updates, the SSH server, the ufw firewall, Docker, the MariaDB database, your website and the nightly backup.
- Nothing gives anybody else access to your computer. The setup only downloads programs from the official places listed.
- Only ports 80 and 443 (Caddy, after the front door step) can be reached from the internet. SSH (22) is for your home network only. The rest are for this computer only.
- The checklist (menu `5`) checks all of this every time you run it.
- To remove it: menu `M`, then `12` (Uninstall). It keeps your photos, database and backups.

Every program the garden setup can install is on this page: what it is, why it is there, and the official page it comes from, so you can read more. Nothing else is installed. The setup also shows each of these on screen, with its commands, before it runs them, and saves everything it prints in `logs/`.

Three groups: - **Necessary.** The website cannot run without these. The install does them. - **Optional.** These are recommended. The install asks yes or no for each one. - **Advanced optional.** Do these later, from the menu (`M` for More).

Nothing gives anybody else access to your computer. Every password is made on your computer or typed by you. The setup sends nothing anywhere: it only downloads programs from the official places listed below.

Necessary

What	What it is	Why	Where it runs	Official page
Ubuntu updates	Ubuntu's own updates through `apt` (its app store)	Safety fixes come first	Ubuntu	https://ubuntu.com/server/docs/package-management

What	What it is	Why	Where it runs	Official page
git, curl, openssl, rsync	Small standard tools	git keeps a history of your changes. curl fetches web pages. openssl makes random passwords. rsync is a copying tool other programs use (you never need to run it yourself).	Ubuntu	https://git-scm.com/doc, https://curl.se/docs/
OpenSSH server	Lets you use this computer's terminal from another computer at home	So you can manage the laptop with the lid closed	Ubuntu, port 22, home network only	https://ubuntu.com/server/docs/openssh-server
ufw firewall	Decides which connections may come in	Blocks everything except SSH from home, and later 80 and 443	Ubuntu	https://ubuntu.com/server/docs/how-to/security/firewalls/
Docker Engine and Compose	Runs each program in its own sealed box (a "container")	Programs can't break Ubuntu or each other. The same setup works on any computer.	Ubuntu	https://docs.docker.com/engine/install/ubuntu/
MariaDB 11	The database. It speaks the same language as MySQL.	Holds plants, comments, logins and the wiki's pages	Docker, inside only (no port)	https://mariadb.com/kb/en/documentation/
Jun's Garden website	Your site, written in Rust, built on your computer from `rust/`	It is your garden: catalogue, gallery, blog, data, guestbook, plant admin, the phone app's server	Docker, `127.0.0.1:8030`	https://www.rust-lang.org/learn, `docs/rust-port/CODE-TOUR.md`
Toolbox	A tiny Alpine Linux container with database and archive tools	Backup, restore and set-password work the same on Ubuntu and Windows	Docker, only while a job runs	https://alpinelinux.org/about/
Nightly backup	One line in your cron (Ubuntu's alarm clock)	A full backup every night at 2:30 am, kept 7 days (with the photos, about 6 GB for all 7)	Ubuntu	https://manpages.ubuntu.com/manpages/noble/man5/crontab.5.html
Swap file (small laptops only)	Disk space used as extra memory	Building the website needs about 3 GB of memory	Ubuntu	https://manpages.ubuntu.com/manpages/noble/man8/swapon.8.html

Optional (the install asks)

What	What it is	Why	Where it runs	Official page
Garden tools (`/beds`)	A small Python program: bed planner, checkup, diagnose, tag scan, spend, wiki queue	They were part of the live garden. The phone app's camera tools use them too.	Docker, `127.0.0.1:8025`	`beds/README.md`, https://docs.python.org/3/
Wiki (off unless you choose it)	Wiki.js 2, a modern wiki. Your garden articles are a separate download, `junsgarden-wiki-addon.zip`; without it the wiki starts empty	Your encyclopedia, yours to edit. Only if you want it: one more program to download, and about 30 minutes to add the pages	Docker, `127.0.0.1:3000`	https://docs.requarks.io, https://js.wiki
AI keys (Gemini, OpenRouter)	Nothing is installed: you paste YOUR own free keys with menu `K`. Gemini (Google AI Studio) reads tag photos and fills plant facts. OpenRouter is only a backup that passes the same request to a free model when Gemini is busy. Both are your own accounts; nobody else can see them, and they never see your computer, only the plant photo or question the garden sends.	So the AI boxes work	Settings only	https://ai.google.dev/gemini-api/docs, https://openrouter.ai/docs, guide 16
Email (Gmail)	Your Gmail sends the site's emails (password links)	Email sent straight from a home connection lands in spam	Settings only, nothing installed	https://support.google.com/accounts/answer/185833
Desktop icon	"Jun's Garden" in the app list and on the desktop	Opens the menu with one click	Ubuntu	https://specifications.freedesktop.org/desktop-entry-spec/latest/
Docker Desktop (Windows only, if you choose it)	Docker Inc.'s Windows app: it runs the garden's programs and shows them with buttons (start, stop, their messages). Installed with winget; free for personal use	The easy way to manage the garden on a Windows PC. The other way, Docker inside Ubuntu, needs nothing more on Windows	Windows	https://docs.docker.com/desktop/setup/install/windows-install/, guide 26

Advanced optional (later, from menu `M`)

What	What it is	Why	Where it runs	Official page
Caddy	A web server that is the front door, with automatic HTTPS	One door with the same gates as the old site. It gets the padlock certificate by itself.	Ubuntu, ports **80 and 443**, the only two the router forwards	https://caddyserver.com/docs/
Authelia	Your sign-in page, at `/auth`	The garden tools open only after you sign in. Wrong passwords lock the name for a while.	Docker, `127.0.0.1:9091`	https://www.authelia.com/overview/prologue/introduction/
No-IP updater (DUC)	Tells No-IP your home internet address every few minutes	Keeps `junsgarden.zapto.org` pointing at your home	Ubuntu	https://www.noip.com/support/knowledgebase/install-linux-3-x-dynamic-update-client-duc
Router settings	You do this in your router: reserve the laptop's address, forward 80 and 443	Visitors from the internet can reach Caddy, and only Caddy	Your router	`START-HERE/14-reserve-your-ip.md`
n8n	Drag-and-drop automations ("soil is dry, so email me")	Fun and powerful when you are ready	Docker, `127.0.0.1:5678`	https://docs.n8n.io
Adminer	A one-page web viewer for the database	Look inside the tables without typing SQL	Docker, `127.0.0.1:8082`, only while you use it	https://www.adminer.org
LiteLLM (menu `M`, `15`)	One private address for AI, with a monthly spending limit	So a paid AI key can never cost more than you chose	Docker, `127.0.0.1:4000`	https://docs.litellm.ai
DuckDNS updater (menu `M`, `13`)	One cron line that tells DuckDNS your home address, for subdomains	Free extra names (wiki., n8n., ...) behind your sign-in page	Ubuntu (cron)	https://www.duckdns.org/install.jsp

What	What it is	Why	Where it runs	Official page
Tailscale (menu `M, 17`)	Tailscale's own program: a private network of only your devices	Reach the garden computer from anywhere, with no port opened in the router	Ubuntu	https://tailscale.com/kb/1017/install
Local AI (Ollama)	AI models on a strong computer at home (not this laptop)	Private, free AI for the site's AI boxes	That other computer	https://github.com/ollama/ollama, `START-HERE/15-local-ai.md`
cifs-utils (NAS)	Ubuntu's program for connecting a network drive (SMB), from Ubuntu's store	A second copy of every backup on your NAS	Ubuntu (menu 6, then 4)	https://manpages.ubuntu.com/manpages/noble/man8/mount.cifs.8.html
rclone (Google Drive)	A well-known free program that copies files to cloud storage, from Ubuntu's store	A weekly copy of your backup away from home (3-2-1)	Ubuntu (menu 6, then 5)	https://rclone.org/drive/
Android Studio (your choice, not installed by the setup)	Google's program for building Android apps	To work on the garden's phone app in `android-app/`	Your desktop	https://developer.android.com/studio, `START-HERE/17-android-app-and-ai-coding.md`

The wiki's one-time setup helper is built here from `docker/wiki/` on the official **python:3.12-alpine** image (https://hub.docker.com/_/python) and is thrown away after it runs.

The doors (ports): what can be reached from where

Port	Program	Who can reach it
22	SSH	Only computers on your home network (firewall rule)
80, 443	Caddy (after the front door step)	The internet. These are the only two ports the router forwards.

Port	Program	Who can reach it
8030, 8025, 3000, 9091, 5678, 8082, 4000	website, garden tools, wiki, sign-in page, n8n, Adminer, LiteLLM	This computer only (`127.0.0.1`). Caddy passes visits to them.
3306	database	Nobody outside Docker

The checklist (menu `5`) checks all of this every time you run it.

How to look at anything yourself

You want to see	Type this in a terminal in the garden folder
Which programs are running	`docker ps`
What a program has been saying	`docker logs --tail 50 garden-site` (or garden-db, junsgarden-ingest, garden-wiki, garden-auth)
Every setting (no passwords)	menu `8` Settings
How each program is started	`cat docker-compose.yml` (every line has a comment)
How each program is built	`cat rust/Dockerfile`, `cat docker/tools/Dockerfile`, `cat beds/Dockerfile`
The front door's settings	`cat /etc/caddy/Caddyfile`
What the setup did and when	the files in `logs/`, and `logs/install-progress.txt`
The firewall rules	`sudo ufw status verbose`
Your nightly backup line	`crontab -l`

Removing it

Menu `M`, then `12` (Uninstall), stops and removes the garden's programs, the nightly backup and the icon, and removes Caddy and the No-IP updater too. It keeps your photos, database and backups. It prints the command that removes Docker, so you decide.